Towards Explainable Graph-Verfied Neuro-symbolic Chest X-Ray Report Generation

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Abstract. Automated radiology report generation remains challenged by factual hallucinations, clinically erroneous assertions that undermine diagnostic reliability. The generated texts need to be verified with concepts and knowledge graph-based methods to ensure explainability. We present a Neuro-symbolic graph-guided Generator (Nesy-Gen) for report generation from Chest X-ray images, that enforces biological plausibility through ante-hoc reasoning to validate assertions before token generation. Nesy-Gen's dual-path architecture integrates a Vision Branch (Swin Transformer with Vision-Entity Cross-Attention) which perceives medical entities, and a neuro-symbolic Branch (PrimeKG traversal via Temporal Steiner Trees and Logic Tensor Networks) which flags claims with satisfiability scores to ensure verifiable provenance for every generated statement. A three-layer consistency detection gate further enforces NLI consistency, visual grounding, and graph reachability. Evaluated on benchmark datasets, Nesy-Gen achieves state-of-the-art performance across multiple metrics on IU X-ray and a 64% relative METEOR improvement on MIMIC-CXR over the strongest baseline and best clinical F1. The code will become publicly available.

Keywords: Radiology report generation · Chest X-ray · Knowledge Graphs · GraphRAG · Neuro-symbolic AI.

1 Introduction

Medical imaging plays a central role in clinical decision-making, and the automated generation of radiology reports [16]. Although deep learning approaches, including conventional encoder-decoder frameworks and large language models [22], have achieved remarkable linguistic fluency, they predominantly function as statistical text completers relying on fast pattern recognition (System 1 thinking). Consequently, these data-driven models are highly susceptible to visual and textual data biases, frequently suffering from factual hallucinations where they generate linguistically coherent but clinically incorrect claims [9]. Recent work has demonstrated that grounding visual features in structured biomedical knowledge graphs, can improve report interpretability [19].

We propose a neuro-symbolic graph-guided Generator (Nesy-Gen), featuring a dual-path architecture. Our approach enforces biological plausibility through active ante-hoc explainability. Our main contributions are: (i) a dual-path framework that combines a Swin Transformer-based vision branch with a graph reasoning branch for clinically grounded and interpretable report generation. (ii) an aggregated consistency mechanism that suppresses statistically plausible but medically invalid token sequences. (iii) On IU X-ray [4] and MIMIC-CXR [10], experimental results and ablations show the advantage of the system modules; qualitative analysis shows the rationales extracted from the learned subgraph.

2 Related Work

Radiology report generation has evolved from traditional CNN-RNN architectures [28, 27] to advanced Transformer and LLM frameworks [25]. Despite their linguistic fluency, these deep learning systems act as statistical text completers and struggle with inherent data biases [20]. Consequently, they frequently generate generic templates or factual hallucinations containing clinically incorrect claims [8]. This fundamental lack of interpretability and reliability remains a major barrier to their safe clinical deployment [8]. To mitigate these issues, recent methods integrate Knowledge Graphs (KGs) [6] for domain supervision, employing shared latent spaces [12], disease-specific GCNs [11] or explicit visual-concept alignment like RadAlign [5]. Our framework fundamentally advances this paradigm by using neuro-symbolic, KG-guided, ante-hoc verification to explicitly reject unsupported claims, ensuring biological plausibility and verifiable provenance for every generated statement before text generation.

3 Methodology

Let $\mathcal{D} = \{(\mathcal{I}_i, \mathcal{T}_i, \mathcal{R}_i)\}_{i=1}^{N}$ denote a dataset of chest X-ray examinations, where $\mathcal{I}_i \in \mathbb{R}^{H\times W}$ is a frontal radiograph, $\mathcal{T}_i$ is the associated clinical indication (e.g., "shortness of breath"), and $\mathcal{R}_i = \{s_1, \ldots, s_{L_i}\}$ is the corresponding radiology report comprising L_i sentences. The framework is depicted in Figure 1. We reformulate the problem as neuro-symbolic constrained generation: find a report $\mathcal{R}$ and an associated neuro-symbolic, sub-graph $\mathcal{G}^* \subset$ PrimeKG such that:

$$\mathcal{R}^* = \arg\max_{\mathcal{R},\mathcal{G}^*} P(\mathcal{R}|\mathcal{I},\mathcal{T},\mathcal{G}^*) \quad \text{s.t.} \quad \varPhi(\mathcal{G}^*,\mathcal{I},\mathcal{T}) = 1, \tag{1}$$

where $\varPhi(\cdot)$ encodes domain constraints: anatomical plausibility (findings localize to correct structures), biological validity (causal paths respect pathophysiology), and temporal consistency (progression follows known trajectories).

A Swin Transformer encodes $\mathcal{I}_{xray}$ into patch features, followed by cross-modal alignment with clinical entities (Module 0) and autoregressive decoding:

$$\mathcal{I}_{xray} \xrightarrow{\text{SwinT}} V_{patches} \in \mathbb{R}^{N\times 768} \xrightarrow{\text{Proj}} V \in \mathbb{R}^{N\times 256} \xrightarrow{\text{VECA}} V'_{aligned} \xrightarrow{\text{Decoder}} \mathbf{o}_t, \tag{2}$$

where $\mathbf{o}_t = (t_{\text{token}}, n_{\text{id}}, s_{\text{ev}}, \varDelta_t)$ is a quadruplet output: the generated token, its mapped PrimeKG node identifier, an evidence score quantifying visual grounding strength, and a temporal progression indicator (stable/worsening/improving).

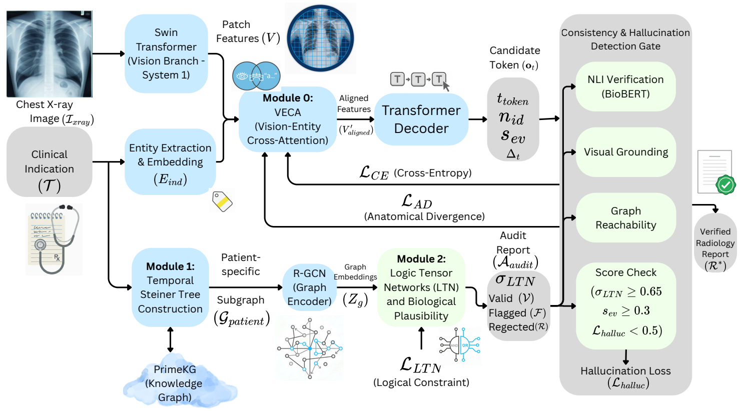

Fig. 1. Dual-path neuro-symbolic architecture for verified X-ray report generation.

A patient-specific subgraph $\mathcal{G}_{patient} \subset$ PrimeKG is extracted via Temporal Steiner Tree (Module 1). Relational Graph Convolutional Networks (R-GCN) [21] encode this graph, handling 30 heterogeneous edge types. Logic Tensor Networks (LTN) then evaluate candidate assertions against the logic constraints:

$$\mathcal{G}_{patient} \xrightarrow{\text{R-GCN}} Z_g \in \mathbb{R}^{|V_g| \times 128} \xrightarrow{\text{LTN}} \mathcal{A}_{audit} = \{\mathcal{V}_{valid}, \mathcal{F}_{flagged}, \mathcal{R}_{rejected}\} \quad (3)$$

The audit report classifies nodes based on satisfiability scores $\sigma_{LTN} \in [0, 1]$: $\mathcal{V}_{valid}$ ($\sigma \geq \beta$), $\mathcal{F}_{flagged}$ ($\gamma \leq \sigma < \beta$), and $\mathcal{R}_{rejected}$ ($\sigma < \gamma$). A Consistency Gate emits a token if and only if:

$$\text{Accept}(t) \iff n_{\text{id}}(t) \in \mathcal{V}_{valid} \wedge s_{\text{ev}}(t) \geq \delta \wedge \mathcal{L}_{halluc}(t) < \epsilon, \quad (4)$$

where β, γ, δ, and ϵ are validation-tuned thresholds detailed in Section 4. We employ a validation-driven grid search that filters hyperparameter candidates by clinical safety constraints before selecting a Pareto-optimal configuration. This staged approach ensures the final model prioritizes clinical faithfulness and logical consistency over raw fluency metrics before proceeding to full-scale training.

3.1 Module 0: Vision-Entity Cross-Attention (VECA)

We parse the indication text using ScispaCy [13] to extract medical entities, embedding them via pre-trained TransE [1], utilized because it efficiently handles large KGs like PrimeKG by modeling triple relations as translations ($\mathbf{h}+\mathbf{r} \approx \mathbf{t}$). To preserve PrimeKG's pre-learned relational structure, the resulting embeddings $E_{ind} = \{e_1, \ldots, e_k\} \in \mathbb{R}^{k \times 256}$ are frozen during training, updating only downstream projection layers. If no entities are mapped, we substitute a randomly initialized NULL embedding $e_\emptyset \in \mathbb{R}^{256}$, which, unlike the frozen TransE vectors, is fully trainable via back-propagation. We compute a soft attention matrix using scaled cosine similarity:

$$S[i,j] = \frac{\exp(\cos(V_i, E_j)/\tau)}{\sum_{j'=1}^{k} \exp(\cos(V_i, E_{j'})/\tau)} \tag{5}$$

We set the temperature τ as 0.07, following RadZero's [15] empirically validated setting, balancing attention sharpness against noise robustness. Then the visual representations are updated as $V_i' = \sum_{j=1}^{k} S[i,j] \cdot V_i$ so that image regions are semantically aligned with the clinical indication (e.g., cardiac structures for "chest pain"), suppressing irrelevant peripheral fields.

3.2 Module 1: Temporal and Causal-related Subgraph Construction

Unstructured report text undergoes negation detection, via NegBio [17, 23] patterns, following concept normalization to UMLS CUIs, we use an offline CUI-to-PrimeKG crosswalk (constructed from PrimeKG node metadata and ontology/UMLS cross-references) to resolve final graph node indices. The output is an initial node set $\mathcal{N}_{extracted}$ with type annotations Disease, Phenotype, Anatomy, Drug, BiologicalProcess. Given terminals $T = \mathcal{N}_{extracted}$ and a directed weighted temporal graph $\mathcal{G}_t = (e, w, \tau_e) = ((h, r, t), w, \tau_e)$, head h, relation r, and tail t, find a minimum-cost tree $\mathcal{T}^* \subset \mathcal{G}_t$ that connects all terminals while respecting temporal ordering and prioritizing related edges. Weighting w of the edge e is:

$$w(e_{ij}) = \alpha \cdot d_{temp}(i,j) + (1-\alpha) \cdot (1 - r_{clin}(i,j)), \tag{6}$$

where $d_{temp}(i,j) = \frac{|\tau_e(i) - \tau_e(j)|}{\max(\tau_e(i), \tau_e(j))}$ is normalized temporal distance, τ_e denotes the temporal attribute function, and $r_{clin}(i,j)$ assigns clinical relevance: 1.0 for 'causes', 0.7 for 'located_in', 0.5 for 'associated_with', 0.3 otherwise[1]. Parameter $\alpha = 0.6$ was tuned on 100 manually-annotated longitudinal cases from Chest ImaGenome paired studies (excluded from final test evaluation) to maximize F1 (0.78) for edge-level accuracy against radiologist-verified causal chains. We approximate the NP-hard Steiner tree problem using the CulT algorithm [18], which identifies minimum-cost paths via Dijkstra's algorithm while strictly filtering for temporal violations and pruning non-essential nodes. The temporal graph used by the Steiner module is constructed from longitudinal study pairs.

To densify sparse longitudinal supervision, we generate synthetic temporal edges using clinical progression templates across entity types (e.g., Finding, Diagnosis). Each template is an ordered tuple $(x_1 \to x_2 \to \cdots)$ with an admissible time window Δt, confidence c_r, and type constraints (e.g., infectious progression: `pneumonia` $\to$ `consolidation` $\to$ `resolution`; or cardiac decompensation: `cardiomegaly` $\to$ `pulmonary edema` $\to$ `pleural effusion`). After linking normalized report entities to PrimeKG, a synthetic edge (h, r, t) is created only if both endpoints match a template, are type-compatible, non-negated, and non-contradictory. Finally, these directed edges enforce Δt temporal constraints and are down-weighted relative to curated edges in the Steiner objective.

[1] The other relations include "interacts", "synergistic interaction", "indication", "contraindication", "drug target", "drug carrier", and "side effect".

3.3 Module 2: Multimodal Constraints

To enforce spatial fidelity, we compare model attention maps A_k with pathology masks M_k for each pathology $k \in \{1, \ldots, K_p\}$, where K_p is the number of pathology concepts extracted from findings/impression and linked to PrimeKG for the current sample. A_k is obtained from the final decoder cross-attention (averaged across heads) over the token span aligned to pathology k, then reshaped to the image grid. M_k is the corresponding anatomical mask: expert Chest ImaGenome annotation when available, otherwise pseudo-mask from CheXpert [7] labels with GradCAM++ [2].

$$W_2(\mathrm{PD}(A_k), \mathrm{PD}(M_k)) = \left(\inf_{\gamma} \sum_{i,j} \gamma_{ij} \left\| (b_i^A, d_i^A) - (b_j^M, d_j^M) \right\|_2^2 \right)^{1/2}, \tag{7}$$

where $\mathrm{PD}(\cdot)$ denotes persistence diagrams (PD_0: connected components, PD_1: holes) [14]. The anatomical loss is:

$$\mathcal{L}_{AD} = \frac{1}{K_p} \sum_{k=1}^{K_p} W_2(\mathrm{PD}(A_k), \mathrm{PD}(M_k)). \tag{8}$$

We enforce differentiable first-order constraints over the inferred Steiner subgraph $\mathcal{T}^*$. Let $E(\mathcal{T}^*)$ denote its edge set, and let each edge be $e = (u, v, r) \in E(\mathcal{T}^*)$, with source node u, target node v, and relation type r. Let $\mathcal{F}$ and $\mathcal{D}$ be finding and diagnosis node sets extracted from the current report.

We define clause set $\mathcal{K} = \{\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3\}$:

$$\mathcal{C}_1 : \ \forall e, \ \mathrm{BioValid}(e) \Rightarrow \big(\mathrm{TypeCompatible}(e) \wedge \mathrm{TemporalOrdered}(e)\big) \tag{9}$$

Meaning: biologically valid edges must satisfy type compatibility and temporal order.

$$\mathcal{C}_2 : \ \forall f \in \mathcal{F}, \ \exists d \in \mathcal{D}, \ \exists p : \ f \rightsquigarrow d \text{ in } \mathcal{T}^* \tag{10}$$

Meaning: each finding must connect to at least one diagnosis through a valid path.

$$\mathcal{C}_3 : \ \forall e = (h, r, t), \ r = \texttt{located\_in} \Rightarrow \big(\mathrm{Type}(h) = \mathrm{Phenotype} \wedge \mathrm{Type}(t) = \mathrm{Anatomy}\big) \tag{11}$$

Meaning: `located_in` is valid only for phenotype→anatomy, where Type() returns the type of the head h or the tail t entity in the KG.

A type-compatibility matrix $\mathbf{C}_{\mathrm{type}} \in \{0, 1\}^{5 \times 5}$ encodes node-type transitions:

$$\mathrm{TypeCompatible}(e) = \mathbf{C}_{\mathrm{type}}[\mathrm{Type}(h), \mathrm{Type}(t)], \tag{12}$$

for edge $e = (h, r, t)$. We also use $\mathrm{BioValid}(e) \in [0, 1]$ (biological plausibility from relation prior and source reliability) and $\mathrm{TemporalOrdered}(e) \in \{0, 1\}$ (whether h precedes t under temporal index τ).

For each clause $\mathcal{C}_k \in \mathcal{K}$ and $k \in \{1,2,3\}$, let $\phi_k(e) \in [0,1]$ be the fuzzy truth value on edge e, and let $c(e) \in [0,1]$ be edge confidence. We compute confidence-weighted clause satisfaction as

$$\sigma_{LTN}(\mathcal{C}_k) = \frac{\sum_{e \in E(\mathcal{T}^*)} c(e)\phi_k(e)}{\sum_{e \in E(\mathcal{T}^*)} c(e) + \varepsilon}, \tag{13}$$

where $\varepsilon > 0$ prevents division by zero. The term $c(e)$ denotes edge confidence, calculated as $c(e) = c_{\text{map}}(e) \cdot \rho(\text{source}(e))$, where $c_{\text{map}}(e) \in [0,1]$ is mapping confidence and $\rho(\text{source(e)}) \in [0,1]$ is source reliability (e.g., curated ontology > computational prediction). Thus, confidence is applied directly inside σ_{LTN}, so uncertain edges have reduced influence on both clause satisfaction and $\mathcal{L}_{LTN}$.

The LTN objective is

$$\mathcal{L}_{LTN} = \sum_{\mathcal{C}_k \in \mathcal{K}} w_k \left(1 - \sigma_{LTN}(\mathcal{C}_k)\right)^2. \tag{14}$$

For downstream verification (Module 3), we use mean logical consistency:

$$\bar{\sigma}_{LTN} = \frac{1}{|\mathcal{K}|} \sum_{\mathcal{C}_k \in \mathcal{K}} \sigma_{LTN}(\mathcal{C}_k). \tag{15}$$

3.4 Module 3: Hallucination Detection and Conflict Resolution

We use a Natural Language Inference (NLI) Verification via BioBERT to compute visual-textual entailment $P(\text{Entail}|p_{visual}, s_i)$. We also use Visual Grounding to map tokens to PrimeKG anatomical region concepts using regional attention α_{loc} (a scaled dot-product attention between the decoder's hidden state and the encoder's spatial region features). Conflict resolution suppresses any token where entailment falls below 0.5 or grounding below 0.3, while reachability failures trigger uncertainty markers. Validated assertions are accepted with a confidence score of $\min(P(\text{Entail}), s_{ground}, \sigma_{LTN})$, where s_{ground} is the visual grounding score computed from cross-attention mass over the expected anatomical region; we compare entailment, grounding, and logical satisfiability because they capture complementary failure modes (textual contradiction, visual mismatch, and biological inconsistency). The hallucination detection head is optimized via binary cross-entropy on validated and synthetically corrupted samples, contributing to the total loss:

$$\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda_1 \mathcal{L}_{AD} + \lambda_2 \mathcal{L}_{LTN} + \lambda_3 \mathcal{L}_{halluc} \tag{16}$$

where $\mathcal{L}_{CE}$ is the cross-entropy loss for autoregressive modeling.

4 Experimental Settings and Results

Experimental Settings. Consistency Gate thresholds were selected by grid search on the validation set: LTN acceptance $\beta = 0.65$, flagging $\gamma = 0.50$, minimum grounding evidence $\delta = 0.30$, and hallucination rejection $\epsilon = 0.50$. For temporal graph construction, we use 500 longitudinal Chest ImaGenome pairs [26] plus rule-based synthetic progression edges, built strictly from non-test data. LTN clause weights were tuned by ablation as $w_{\mathcal{C}_1} = 1.2$, $w_{\mathcal{C}_2} = 1.0$, and $w_{\mathcal{C}_3} = 1.2$; higher weights on $\mathcal{C}_1$ and $\mathcal{C}_3$ emphasize type compatibility and anatomical localization, the most common biological failure modes. In Eq. 16, we set $\lambda_1 = 0.3$, $\lambda_2 = 0.2$, $\lambda_3 = 0.1$. Training uses AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}), with 5-epoch linear warmup ($10^{-7} \to 5 \times 10^{-5}$) followed by cosine decay to 10^{-6} over 25 epochs. We evaluate Nesy-Gen on IU X-ray and MIMIC-CXR, using the 7:1:2 IU split as in [3] and the official MIMIC split; Chest ImaGenome gold-standard patients are withheld to avoid leakage [26]. We report BLEU, ROUGE-L, METEOR, and entity-level clinical correctness using Precision (P), Recall (R), and F1. For each predicted/reference report pair, medical entities are extracted and linked to KG nodes to obtain $\hat{\mathcal{E}}$ and $\mathcal{E}$. We compute $P = \frac{|\hat{\mathcal{E}} \cap \mathcal{E}|}{|\hat{\mathcal{E}}|}$, $R = \frac{|\hat{\mathcal{E}} \cap \mathcal{E}|}{|\mathcal{E}|}$, and $F1 = \frac{2PR}{P+R}$, then average per-sample scores over the test set to measure clinical concept recovery beyond surface fluency.

Table 1. Performance comparison with ablation study on IU X-ray and MIMIC-CXR.

Model / Variant	BL1	BL2	BL3	BL4	MR	RGL	P	R	F1
IU X-ray Dataset									
State-of-the-Art									
KGAE [12]	41.7	26.3	1.81	12.6	14.9	31.8	-	-	-
TRRG [24]	48.2	30.2	21.7	15.1	**20.9**	37.7	-	-	-
R2Gen [3]	47.0	30.4	21.9	16.5	18.7	37.1	-	-	-
Ablation Study									
Baseline (Swin+Trans)	39.2	24.5	17.8	11.2	15.5	31.2	33.8	27.4	30.3
+ VECA	44.1	29.8	21.1	15.4	18.8	34.5	37.1	31.8	34.2
+ Temporal ST	46.5	31.2	22.4	16.8	19.9	36.8	39.4	34.2	36.6
+ LTN	47.8	32.4	23.5	17.9	20.1	37.2	40.6	35.6	37.9
Ours (Full)	**48.8**	**33.4**	**24.1**	**18.5**	20.8	**37.9**	**41.2**	**36.5**	**38.7**
MIMIC-CXR Dataset									
State-of-the-Art									
KGAE [12]	22.1	14.4	9.6	6.2	9.7	20.8	-	-	-
TRRG [24]	**43.6**	**29.8**	**21.3**	**15.7**	16.7	**33.6**	40.3	**39.9**	39.3
R2Gen [3]	35.3	21.8	14.5	10.3	14.2	27.7	33.3	27.3	27.6
Ablation Study									
Baseline (Swin+Trans)	28.5	15.2	8.8	5.4	11.8	22.4	30.1	24.0	26.7
+ VECA	33.2	19.4	12.6	8.2	14.5	27.2	36.2	29.1	32.3
+ Temporal ST	35.8	21.0	13.8	9.9	22.0	28.5	41.0	33.4	36.8
+ LTN	36.8	21.8	14.4	10.2	26.8	29.4	44.7	35.9	39.8
Ours (Full)	37.5	22.8	14.1	9.8	**27.4**	30.2	**48.5**	38.8	**43.1**

Results. As shown in Table 1, Nesy-Gen achieves state-of-the-art performance on IU X-ray and a strong METEOR lead on MIMIC-CXR dataset. Ablations show complementary gains: VECA improves vision–indication alignment, Tem-

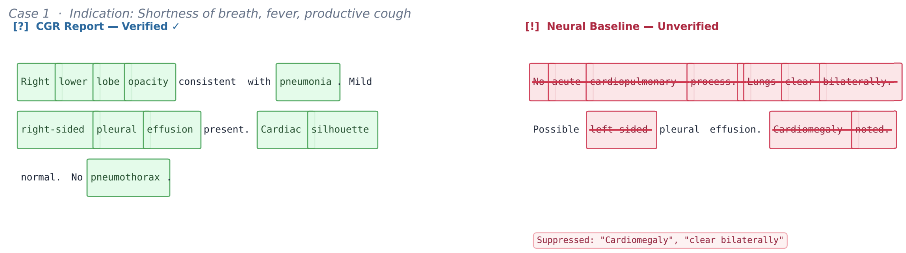

Fig. 2. Token-level comparison of Nesy-Gen (left) and R2Gen [3] (right) on IU X-ray **[?]**: flagged ($0.5 \leq \sigma_{LTN} < 0.65$); **[!]**: uncertain reachability. Nesy-Gen suppresses baseline-only errors such as *cardiomegaly* ($s_{ground} = 0.18$) and *clear bilaterally* (no $\mathcal{T}^*$ path to `Normal Lung`).

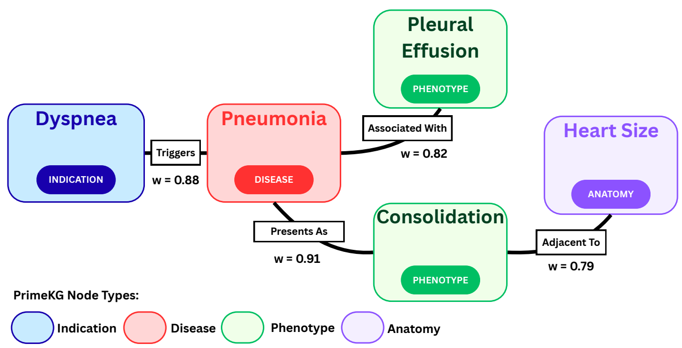

Fig. 3. Temporal Steiner Subgraph extracted from PrimeKG for Dyspnea (IU-Xray).

poral Steiner improves temporal coherence, and LTN gives the largest single gain on MIMIC by enforcing biological plausibility during training.

Qualitative Analysis. In Fig. 2, token-level verification (entailment $P(\text{Entail}) \geq 0.5$, grounding $s_{\text{ground}} \geq 0.3$, and graph reachability in $\mathcal{T}^*$) retains clinically supported findings (e.g., right lower lobe opacity, pleural effusion) and suppresses unsupported tokens (e.g., cardiomegaly, clear bilaterally). Fig. 3 shows the corresponding PrimeKG subgraph (node types, relation reliability, edge confidence).

5 Conclusion

Explainability based on Knowledge Graphs is critical for the cross-modality task of automated radiology report generation. We proposed Nesy-Gen, a neuro-symbolic framework that combines a Vision Branch with a Neuro-symbolic Branch. By using graph verification before the text is generated, the system ensures that statements are medically plausible w.r.t the learned sub-KG. Future work will focus on clinical experts' validation of the explainable graph paths.

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